



Sample Application for Research Training and Career Development Funding

Through the F99/K00 Transition to Aging Research for Predoctoral Students Awards, NIA supports exceptional graduate students in finishing their dissertation research on any topic and continuing to a postdoctoral position conducting research on aging. Each award has two phases: the F99 phase supporting predoctoral training and the K00 phase supporting postdoctoral training.

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PROGRAM CONTACT:
Dana Plude
[REDACTED]

SUMMARY STATEMENT
(Privileged Communication)

Release Date: 03/31/2021
Revised Date:

YU,KEXIN
University of Southern California
[REDACTED]

Application Number: 1 F99 AG068492-01A1
Formerly: 1F99AG068492-01

Review Group: ZAG1 ZIJ-D (M1)
National Institute on Aging Special Emphasis Panel
Transition to Aging
Meeting Date: 02/17/2021
Council: MAY 2021
Requested Start: 07/01/2021
PCC: 2BBCHDP

Project Title: Examining the Longitudinal Influence of the Physical and Social
Environments on Social Isolation and Cognitive Health:
contextualizing the role of technology

Requested:

Sponsor:
Department: Social Work
Organization: UNIVERSITY OF SOUTHERN CALIFORNIA
City, State: LOS ANGELES CALIFORNIA

SRG Action: Impact Score:19
Next Steps: Visit https://grants.nih.gov/grants/next_steps.htm
Human Subjects: 30-Human subjects involved - Certified, no SRG concerns
Animal Subjects: 10-No live vertebrate animals involved for competing appl.
Gender: 1A-Both genders, scientifically acceptable
Minority: 1A-Minorities and non-minorities, scientifically acceptable
Age: 8A-Only Older Adults, scientifically acceptable

RESUME AND SUMMARY OF DISCUSSION: This resubmission application, for a Transition to Aging Research for Predoctoral Students (F99/K00) award, is tendered by the University of Southern California on behalf of Principal Investigator (PI) Ms. Kexin Yu. It seeks five years of funding to investigate the influences of living environments on social isolation and cognitive health in older adults. To address this, the PI will a) examine the effects of information and communication technology use, within physical and social environments, on cognitive decline and social isolation using data from the National Health and Aging Trends Study (NHATS) (F99 phase); and b) develop a technology-facilitated behavioral health intervention to reduce social isolation and prevent cognitive decline employing a community-based participatory design approach (K00 phase). This work builds on Ms. Yu's interests and experience in research on the association of social isolation, loneliness, and cognitive decline. If successful, this work has the potential to generate novel insights regarding environmental factors that increase the risk of social isolation and potential effects on cognitive health in older adults. Other notable strengths include the applicant's comprehensive responses to initial reviews, increased productivity since the initial submission, multiple publications, and excellent training plan. In addition, the panel felt that the applicant's mentors for both phases are strong in terms of mentorship. However, multiple weaknesses temper enthusiasm, including the applicant's relative lack of experience in primary data collection (e.g., structured interviews), the need for additional explanation regarding interpretation of data, lack of clarity regarding how the sample sizes were determined, and the training plan's lack of consideration for how to address pitfalls. This resubmission application has provided thorough and satisfactory responses to previous critiques.

TRAINING IN THE RESPONSIBLE CONDUCT OF RESEARCH: Acceptable. The applicant has addressed all required components.

DESCRIPTION (provided by applicant): Social wellbeing and cognitive health are two pillars of successful aging, and they are interrelated. Three decades of research have consistently shown that social isolation is associated with declines in cognitive functioning. Although existing literature has explored the individual-level risk and protective factors, researchers' attention has been less paid to the extent to which physical and social characteristics of the living environment affect social isolation and cognitive health. The Ecological Theory of Aging (ETA) posits environmental factors have significant influences on health and social wellbeing. Negative physical characteristics of the environment were hypothesized to be related to increased social isolation and cognitive decline longitudinally. An increasing amount of research investigated the association between ICT use and social isolation, yet whether ICT use would interact with the effect of environmental factors on the risk of social isolation has not yet been empirically tested. Besides, the influence of ICT use on cognitive health has not been well understood. Promoting social interaction using technology-based approaches have the potential to alleviate social isolation among older people and prevent cognitive decline. The proposed dissertation work (F99) will examine the effect of physical and social environments of living on social isolation and cognitive health. The effects of ICT use will be investigated within living contexts. Multi-level modeling and structural equation modeling methods will be employed to conduct secondary data analysis using data from the National Health & Aging Trend Study (NHATS) wave 5 to 9. The postdoctoral stage (K00) is guided by the overarching aim of building knowledge on the behavioral and contextual pathways that could explain the association between social isolation and cognitive decline and developing a community-based behavioral health intervention. At the K00 phase, the PI will increase understanding of the social isolation and cognitive health research landscape by conducting a systematic review. Longitudinal data from the ongoing Internet-based Conversational Engagement Clinical Trial (I-CONNECT) will be analyzed to examine the association between social isolation and change in neuropsychological outcomes. The PI will utilize the longitudinal data collected with the Collaborative Aging Research Using Technology (CART) platform to investigate the relationships between daily activities, social engagement, and the cognitive health of older participants. Knowledge to be built at F99 and K00 stages will be synthesized to inform the participatory co-design of a technology-facilitated intervention program for social isolation and cognitive decline. The proposed research will produce

knowledge on the influence of environmental factors on social isolation and cognitive health and achieve the PI's long-term professional goal of developing behavioral health intervention to address social isolation and prevent the onset of Alzheimer's Disease and Related Dementias.

PUBLIC HEALTH RELEVANCE: The proposed dissertation work will contribute to the fundamental knowledge with longitudinal empirical evidence on the influence of physical, social environmental factors on social isolation and cognitive health, and contextualized the role of technology use. The postdoctoral research will build knowledge on the behavioral and contextual pathways that could explain the association between social isolation and cognitive decline in later life. Knowledge to be built in the predoctoral and postdoctoral stages will be synthesized to inform the development of a technology-facilitated community-based intervention program to reduce social isolation and prevent cognitive decline among older adults.

DISCLAIMER: Please note that the following critiques were prepared by the reviewers prior to the Study Section meeting and are provided in an essentially unedited form. While there is opportunity for the reviewers to update or revise their written evaluation, based upon the group's discussion, there is no guarantee that individual critiques have been updated subsequent to the discussion at the meeting. Therefore, the critiques may not fully reflect the final opinions of the individual reviewers at the close of group discussion or the final majority opinion of the group. Thus, the Resume and Summary of Discussion is the final word on what the reviewers actually considered critical at the meeting.

CRITIQUE 1:

Fellowship Applicant: 3
Sponsors, Collaborators, and Consultants: 1
Research Training Plan: 1
Training Potential: 2
Institutional Environment & Commitment to Training: 1

Overall Impact:

This is a resubmission reflecting careful attention to weaknesses noted with the initial submission. This application details the research and training activities in support of Ms. Yu's work as a PhD student in social work at the University of Southern California (USC). She has planned a research career focused on understanding environmental factors related to social isolation and cognitive decline among older adults and plans to develop technology-facilitated behavioral health interventions for mitigating social isolation and cognitive decline. The proposed research aims focus on a) examining the effects of information and communication technology use, within physical and social environments, on cognitive decline and social isolation using data from the National Health and Aging Trends Study (NHATS) (F99 phase); and b) developing a technology-facilitated behavioral health intervention to reduce social isolation and prevent cognitive decline employing a community-based participatory design approach (K00 phase). The applicant is clearly motivated and passionate about her planned research. The proposed research activities and training plans are well-developed and will be facilitated by an outstanding mentoring/support team with strong track records of research funding and junior scholar mentorship. The research studies are well-reasoned and logical for accomplishing the training aims and stand to make novel contributions to prevention research on older adults' social circumstances and cognitive health.

1. Fellowship Applicant: Strengths

- The applicant is pursuing a PhD in social work, exhibits strong academic performance, has enthusiastic letters of support, and provides evidence of scientific productivity as an emerging independent researcher.
- The training goals are clear and well-reasoned.

Weaknesses

- None noted.

2. Sponsors, Collaborators, and Consultants:

Strengths

- The sponsors and collaborators are a strong interdisciplinary team of clinical, biomedical, and translational researchers/scientists with established track records of mentoring and launching independent researchers.
- The applicant's existing collaboration with the K00 phase sponsor (Dr. Hiroko Dodge) and associated collaborators suggests that the transition to the postdoctoral phase will be smooth.
- Enhancements to the team appear appropriate given previous reviews.

Weaknesses

- None noted.

3. Research Training Plan:

Strengths

- Enhancements to the training plan in this resubmission, including clinical trial methods and experience, grant writing, and methods for neuroimaging and statistics, are noted and strong.
- The training plans for both F99 and K00 phases are thorough and well-structured to support the applicant's aims.

Weaknesses

- None noted.

4. Training Potential:

Strengths

- The applicant is clearly motivated to pursue this work.
- Enhancement of the applicant's track record of productivity in support of this resubmission suggest good training potential.

Weaknesses

- None noted.

5. Institutional Environment & Commitment to Training:

Strengths

- The environments for both the F99 and K00 phases are strong.

Weaknesses

- None noted.

Protections for Human Subjects:

Acceptable Risks and Adequate Protections.

Inclusion Plans:

- Sex/Gender: Distribution justified scientifically.
- Race/Ethnicity: Distribution justified scientifically.
- Inclusion/Exclusion Based on Age: Distribution justified scientifically

Resubmission:

- The introduction to the resubmission is thorough and details what appears to be thoughtful and comprehensive inclusions based on initial commentary.

Training in the Responsible Conduct of Research:

Acceptable.

Resource Sharing Plans:

Not Applicable.

Budget and Period of Support:

Recommend as Requested.

CRITIQUE 2:

Fellowship Applicant: 2

Sponsors, Collaborators, and Consultants: 3

Research Training Plan: 3

Training Potential: 2

Institutional Environment & Commitment to Training: 1

Overall Impact:

The applicant is a doctoral student enrolled in the social work program at USC. She has received her master's in social work from the same institution. The applicant's long-term career goal is to develop and employ technology-assisted behavioral interventions to reduce social isolation and prevent cognitive decline. Specifically, in the short-term, she plans to examine the physical and social living environments on older adults' social and cognitive health. This focus stems, firstly, from her previous work examining longitudinal relationships between internet use and health conditions, and loneliness and secondly, from her fieldwork at affordable senior housing communities. The resubmission application has been strengthened and addresses the comments raised by the reviewers. The overall training environment, training/mentoring plan, and the track record of the applicant will ensure the success of the applicant in the proposed research. Enthusiasm for this application is somewhat reduced by the weakness pertaining to the research strategy and still some areas of training that might be lacking considering the ambitious undertaking of the application, especially in the K00 phase.

1. Fellowship Applicant:

Strengths

- The applicant has shown academic and research rigor during her doctoral work at USC, with five published manuscripts (two of them in high-impact factor journals) and 17 abstract presentations.
- The applicant has built an excellent track record during her doctoral training at USC. She has been awarded a USC predoctoral fellowship and Clinical and Translational Science Institute pilot funding, and also won the Nobuo Maeda International Research Award by American Public Health Association.
- The applicant has excellent reference and sponsor letters. They discuss her passion in addressing social isolation and loneliness in older adults, her ability to be productive under high stress, as well as her independent critical thinking, and learning advanced analytical skills.

Weaknesses

- None noted.

2. Sponsors, Collaborators, and Consultants:

Strengths

- The applicant has taken feedback provided by the reviewers in the prior submission to strengthen the team of sponsors and collaborators to achieve her four specific training goals.

- She now has a team of six mentors, three with expertise in social work from USC, and three with expertise in neurology/cognitive neuroscience from Oregon Health & Science University (OHSU). The applicant has worked with all six mentors to develop her research and training plan. In addition, there are two other collaborators added - Dr. Judy Pa (with a clinical background) and Dr. John Ringman (with neuroimaging background).
- All the sponsors bring different aspects of expertise, which will be a tremendous asset for the applicant.
- All the sponsors/co-sponsors have experience with mentoring doctoral and postdoctoral and doctoral trainees.

Weaknesses

- [REDACTED]

3. Research Training Plan:

Strengths

- The research addresses a significant and understudied area of examining how the physical living and social environment affects social isolation and cognitive health.
- The large data set that will be used for analysis from the NHATS.
- The K00 phase will result in a technology-facilitated behavioral health intervention to address social isolation and cognitive decline.

Weaknesses

- Aim 1 has nine hypotheses, all related to secondary data analysis. Although the applicant has some experience with data collection and conducting interviews, that lack of extensive training in the F99 phase is a weakness.
- There is very little information on interpretation of findings from the elaborate nine hypotheses for Aim 1 and discussion of expected versus unexpected findings and potential problems and alternative solutions to address this.
- Although the plan for the K00 phase is well-laid-out, the applicant's main goal is to develop a behavioral intervention; there is very little mention of the type of intervention and how/what technological devices the applicant is thinking of employing to the Collaborative Aging Research Using Technology initiative for the intervention.
- The applicant might not have sufficient training in mhealth platforms and knowledge for development and implementation of information and communication technology for the behavioral intervention. Especially at OHSU, the mentoring team does not include anyone with this expertise.
- A total of 32 participants will be enrolled for the structured interviews. It is not clear how this sample size was arrived at and if n=8 in each group would be a sufficient sample.

4. Training Potential:

Strengths

- The applicant demonstrates significant training potential and the training plan put together by her, in collaboration with the sponsors, is very strong.
- Dr. Chi will serve as the applicant's primary mentor for the F99 phase and has expertise in working design and implementation of community-based interventions in older adults. Dr. Chi has closely worked with the applicant for formulating the research question and experimental design. Dr. Wu from social work at USC has expertise in technology-delivered health interventions and has worked with the applicant as a co-mentor and co-chair of her dissertation. Dr. Lawrence Palinkas is a Professor of Social Policy and Health at USC and will contribute his expertise in mixed-methods and qualitative research design.
- Dr. Dodge is Professor of Neurology at OHSU will serve as the primary sponsor for the K00 phase and provide training in developing clinical trial designs and clinical trial outcomes and

analysis of longitudinal biomarker data. Dr. Jeffrey Kaye will serve as a clinical Alzheimer's disease (AD) mentor and assist to explore technology-assisted AD interventions. Dr. Lisa Silbert at OHSU will assist with training pertaining to neuroimaging methodology and examining brain biomarker outcomes.

- In addition, per feedback provided by previous reviewers, the applicant has added two other collaborators with whom she will work. Dr. Pa, in the F99 phase, to assist with neuroimaging outcomes and gain clinical experience by working with Dr. Ringman in neurology clinics at USC.

Weaknesses

- The applicant might benefit from some didactic coursework on cognitive neuroscience and neuroimaging analysis.

5. Institutional Environment & Commitment to Training:

Strengths

- The environment proposed at both institutions (USC School of Social Work for the F99 phase and OHSU for the K00 phase) are excellent. The Suzanne Dworak-Peck School of Social Work at USC is highly ranked in the nation among the schools of social work. At USC, the applicant will also have access to the Hamovitch Center for Science in the Human Services Core, the Roybal Institute on Aging, and the USC AD Research Center.
- The School of Social Work at USC has a strong doctoral program, which includes didactic coursework training, seminars and training workshops for application writing, funding opportunities, and research preparation. The letter of institutional support provided by Dr. Michael Hurlburt, Director of the Doctoral Program for the School of Social Work, is very detailed and lays out precisely the doctoral program requirements, highlighting the strength of the training environment and commitment to training.
- The environment for the K00 training at OHSU School of Medicine is also excellent and the applicant will have access to the Oregon Layton Aging and AD Center.

Weaknesses

- None noted.

Protections for Human Subjects:

Acceptable Risks and Adequate Protections.

Inclusion Plans:

- Sex/Gender: Distribution justified scientifically.
- Race/Ethnicity: Distribution justified scientifically.
- Inclusion/Exclusion Based on Age: Distribution justified scientifically.

Resubmission:

- The applicant has responded well to the reviewers' comments about the training plan and added co-sponsors and included relevant clinical training and learning of neuroimaging data processing and biomarker analysis.

Training in the Responsible Conduct of Research:

Acceptable.

Resource Sharing Plans:

Not Applicable.

CRITIQUE 3:

Fellowship Applicant: 1

Sponsors, Collaborators, and Consultants: 2
Research Training Plan: 2
Training Potential: 1
Institutional Environment & Commitment to Training: 1

Overall Impact:

This is an exemplary application from a very strong and committed applicant who was very responsive to the prior reviews. The letters are strong, as is prior training, publication record, and commitment to aging. The applicant's mentoring team is from multiple institutions for the varied phases of training, covering several disciplines (e.g., aging, anthropology, engineering), which will likely diversify training experiences. The collaborators/sponsors are experienced and established, have funding, and show commitments to mentoring. Weaknesses include the need to further synthesize and detail respective training from the many members of the training team, as well as to more clearly discuss troubleshooting of potential pitfalls, such that a deeper level of critical thinking is demonstrated. Potential issues that could arise, and how to solve them, should be more fully considered.

1. Fellowship Applicant:

Strengths

- The prior concern regarding a lack of publication record has been addressed and is no longer a concern. Congratulations to the applicant for publishing three first-author manuscripts since the application's last submission.
- The applicant has received research accolades/awards, and pilot funding.
- The applicant has completed all program PhD milestones with the exception of her dissertation.
- Strong prior training.
- Strong letters.
- Commitment to social work science is demonstrated, especially as related to aging and physical/social living environments and cognitive aging.

Weaknesses

- None noted.

2. Sponsors, Collaborators, and Consultants:

Strengths

- The applicant's mentoring team is across multiple institutions for the varied phases of training, and they cover several related disciplines (e.g., aging, anthropology, engineering), which can collectively nicely diversify training experiences.
- The collaborators/sponsors are experienced and established and show commitments to mentoring.
- Drs. Silbert and Kaye at OHSU are now co-sponsors. These scientists specialize in neuroimaging and technology to promote positive aging. This addresses concerns from the prior review.
- The sponsor team coordination plan is described, and it should suit the applicant well; however, as detailed below, more information would have helped contextualize relationships amongst the roles.
- Abundant research funding is available from sponsors.

Weaknesses

- The team of sponsor, co-sponsors, and collaborators is very large; however, the inclusion of each individual makes sense and together they have a truly impressive record of mentoring, and the rationale is provided, so this concern does not considerably weaken enthusiasm. The one note regarding this that is of bigger concern, is that while there is some discussion of meetings with particular sponsors, co-sponsors, and collaborators, it is not clear if or when the entire team of mentors will meet with the applicant all together (virtually or in person), and for which milestone.

3. Research Training Plan:

Strengths

- The training plan has been more developed in response to the prior concern that there was not enough detail in it. The aims and goals of training have now been additionally detailed in Table 1. This Training Activities and Timeline table is excellent; it details the training goals, the sponsor/mentor, and the years performed.
- The relationships and impacts of cognitive health and social well-being on aging are critically important research areas. The points of social isolation and related impacts on cognitive aging are especially poignant.
- Regarding planning for the challenges that can arise in clinical research, the applicant plans to receive specific training in clinical trial interventions from her sponsor team, which has expertise in clinical trial research. As mentioned below, however, more detail would have been useful to show critical thinking from her in this realm.
- There was a prior concern that training was lacking in the clinical domain of AD. Now, clinical internships are proposed in neurology clinics, with Dr. Ringman, in the F99 phase of training at USC, and with Dr. Kaye during the K00 phase at OHSU. This is a marked new strength of the training plan, as it is expected that this breadth of clinical experience will help inform research directions and putative interventions with significant clinical impact.
- The conceptual framework of the research in Figure 1 is very useful. The unique portions of each phase are clearly noted and important. The overlapping portions show the logical progression and transition of training and research.
- The inclusion of teaching opportunities is still in the application, and this is a very important strength often overlooked.
- Fascinating topic.
- Wonderfully written application overall.

Weaknesses

- Referencing the need to synthesize and further detail respective training from the sponsors/collaborator, Dr. Kaye will be the primary mentor for assessing the association between social engagement and cognitive health. As the co-sponsor, Dr. Dodge will be the primary mentor for the K00 stage. The applicant plans to move to OHSU for the postdoctoral portion of the training plan. The synthesis of the many others on the team is still unclear.
- Discussion and troubleshooting of specific potential pitfalls would have shown a deeper level of critical thinking regarding potential issues that could arise. What issues could arise, and how to solve them, appears only on the surface and is not fully considered.

4. Training Potential:

Strengths

- The applicant shows increased productivity given recent publications (since the last applicant submission in fact) and this shows dedication to research and its dissemination to enrich the field.
- The applicant is clearly committed to aging research and improving the lives of older individuals through social work-related questions, such as social well-being, social isolation, and cognitive aging.
- Supportive letters indicate overall enthusiasm and support for the applicant, noting a motivated, smart applicant with clinical and research skill sets.
- The clearly outlined training goals and activities show that the applicant and the sponsor are thinking deeply about how to maximize training for the applicant, and how to impact older populations.
- There appears to be a stepwise training plan now, which should yield excellent success for the applicant.

Weaknesses

- None noted.

**5. Institutional Environment & Commitment to Training:
Strengths**

- Both USC and OHSU show commitment to the applicant. Both are strong and excellent overall.

Weaknesses

- None noted.

Protections for Human Subjects:

Acceptable Risks and Adequate Protections.

Inclusion Plans:

- Sex/Gender: Distribution justified scientifically.
- Race/Ethnicity: Distribution justified scientifically.
- Inclusion/Exclusion Based on Age: Distribution justified scientifically.

Training in the Responsible Conduct of Research:

Acceptable.

Budget and Period of Support:

Recommend as Requested.

THE FOLLOWING SECTIONS WERE PREPARED BY THE SCIENTIFIC REVIEW OFFICER TO SUMMARIZE THE OUTCOME OF DISCUSSIONS OF THE REVIEW COMMITTEE, OR REVIEWERS' WRITTEN CRITIQUES, ON THE FOLLOWING ISSUES:

PROTECTION OF HUMAN SUBJECTS: ACCEPTABLE. Acceptable Risks and/or Adequate Protections are described in the application.

INCLUSION OF WOMEN PLAN: ACCEPTABLE. Of the 8,334 human subjects, 52% are women.

INCLUSION OF MINORITIES PLAN: ACCEPTABLE. Of the 8,334 human subjects, at least 21% are underrepresented minorities.

INCLUSION ACROSS THE LIFESPAN: ACCEPTABLE. The team does not anticipate that any participants will be under the age of 18.

COMMITTEE BUDGET RECOMMENDATIONS: The Budget was recommended as requested.

Footnotes for 1 F99 AG068492-01A1; PI Name: Yu, Kexin

NIH has modified its policy regarding the receipt of resubmissions (amended applications). See Guide Notice NOT-OD-18-197 at <https://grants.nih.gov/grants/guide/notice-files/NOT-OD-18-197.html>. The impact/priority score is calculated after discussion of an application by averaging the overall scores (1-9) given by all voting reviewers on the committee and multiplying by 10. The criterion scores are submitted prior to the meeting by the individual reviewers assigned to an application, and are not discussed specifically at the review meeting

or calculated into the overall impact score. Some applications also receive a percentile ranking. For details on the review process, see http://grants.nih.gov/grants/peer_review_process.htm#scoring.